

Centralized RADIUS Authentication

Identity Services Lab 3 - WPA2-Enterprise Wireless Access

PROJECT HIGHLIGHTS

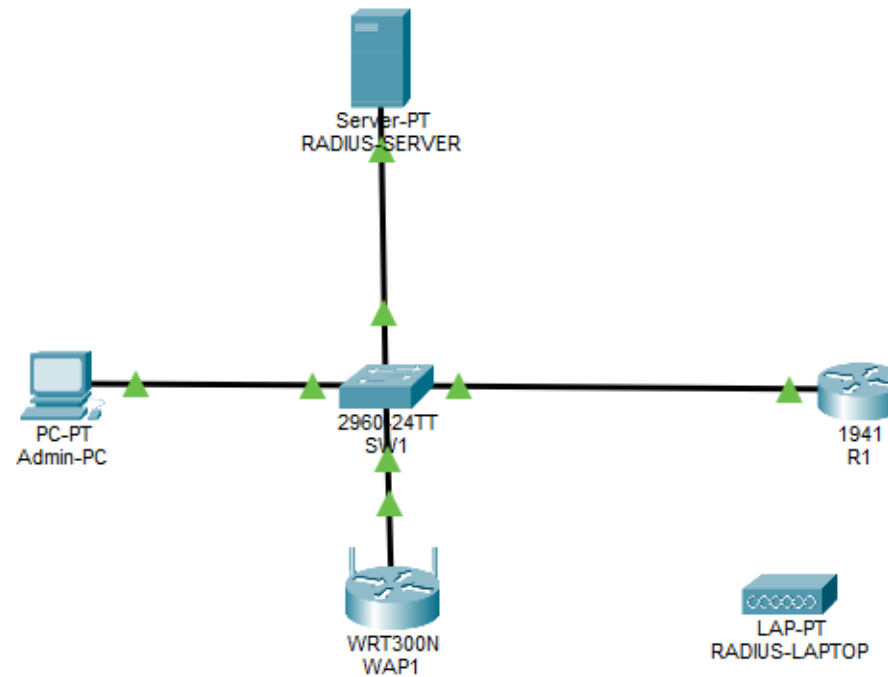
- Centralized user authentication through a RADIUS server
- WPA2-Enterprise wireless access using PEAP
- Successful and rejected identity validation tests
- End-to-end connectivity across the identity services network

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Centralized RADIUS Wireless Topology

VERIFIED



PROJECT EVIDENCE

Designed a centralized identity services network with a RADIUS server, router, switch, wireless access point, administrator workstation, and wireless client.

WHAT THIS DEMONSTRATES

- Identity services architecture
- Layer 2 and Layer 3 connectivity
- Wireless access design

RADIUS Clients and Centralized User

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	Client Name	Client IP	Server Type
1	R1	192.168.10.1	Radius
2	SW1	192.168.10.2	Radius

User Setup

Username

	Username
1	radiusadmin

PROJECT EVIDENCE

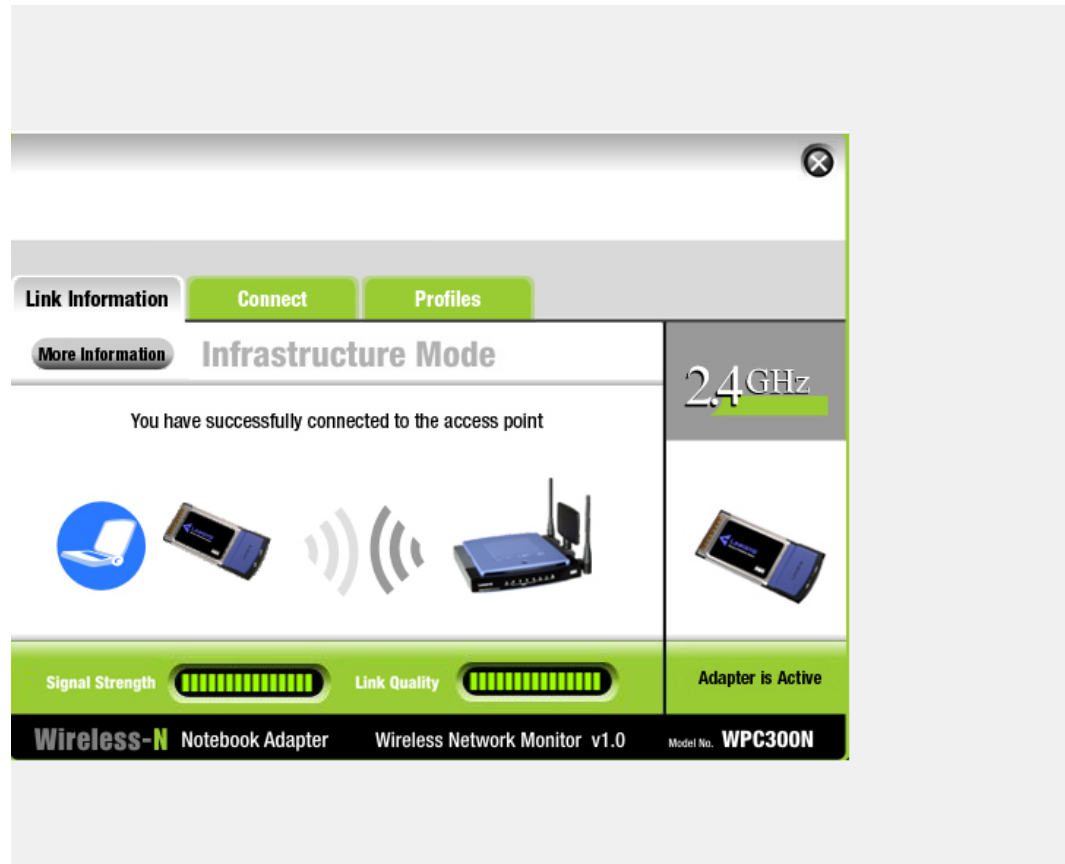
Configured the AAA service with network clients and a centralized user account. Credential and shared-secret fields were excluded from the portfolio image.

WHAT THIS DEMONSTRATES

- Centralized identity management
- RADIUS client registration
- Secure evidence handling

Successful WPA2-Enterprise Association

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PROJECT EVIDENCE

Authenticated the wireless laptop through WPA2-Enterprise and the centralized RADIUS account. The adapter became active and associated with the access point.

WHAT THIS DEMONSTRATES

- WPA2-Enterprise
- PEAP authentication
- Centralized access control

RADIUS Wireless Client Connectivity

VERIFIED

```
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=24ms TTL=255
Reply from 192.168.10.3: bytes=32 time=14ms TTL=255
Reply from 192.168.10.3: bytes=32 time=13ms TTL=255
Reply from 192.168.10.3: bytes=32 time=13ms TTL=255

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 24ms, Average = 16ms

C:\>ping 192.168.10.100

Pinging 192.168.10.100 with 32 bytes of data:

Reply from 192.168.10.100: bytes=32 time=32ms TTL=128
Reply from 192.168.10.100: bytes=32 time=17ms TTL=128
Reply from 192.168.10.100: bytes=32 time=31ms TTL=128
Reply from 192.168.10.100: bytes=32 time=25ms TTL=128

Ping statistics for 192.168.10.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 32ms, Average = 26ms

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=43ms TTL=255
Reply from 192.168.10.1: bytes=32 time=22ms TTL=255
Reply from 192.168.10.1: bytes=32 time=8ms TTL=255
Reply from 192.168.10.1: bytes=32 time=32ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 43ms, Average = 26ms

C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time=33ms TTL=128
Reply from 192.168.10.10: bytes=32 time=23ms TTL=128
Reply from 192.168.10.10: bytes=32 time=27ms TTL=128
Reply from 192.168.10.10: bytes=32 time=37ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 23ms, Maximum = 37ms, Average = 30ms
```

PROJECT EVIDENCE

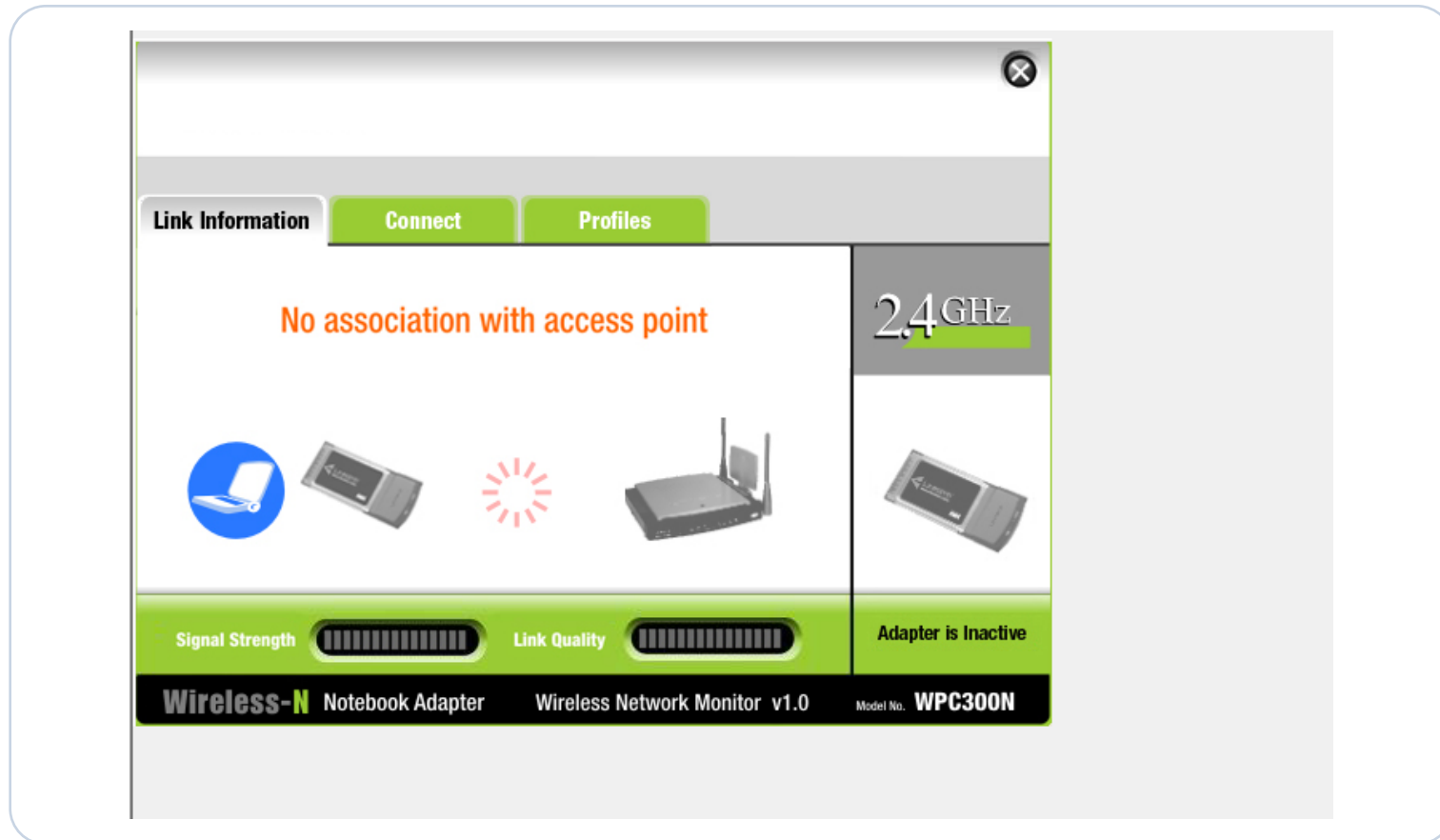
Verified zero-loss connectivity from the authenticated wireless client to WAP1, the RADIUS server, R1, and Admin-PC.

WHAT THIS DEMONSTRATES

- End-to-end ping testing
- IPv4 connectivity validation
- Authenticated network access

Unauthorized RADIUS Authentication Rejected

ACCESS DENIED



PROJECT EVIDENCE

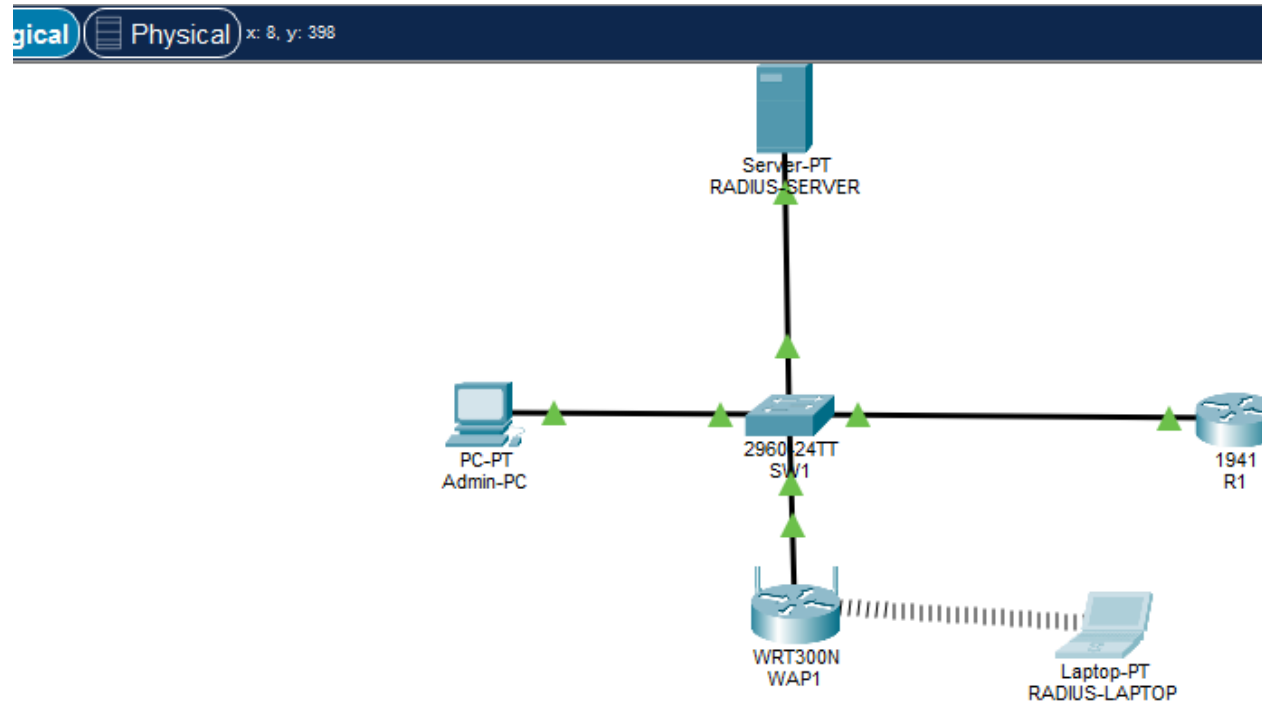
Tested an unauthorized wireless profile and confirmed that the client could not associate with the access point.

WHAT THIS DEMONSTRATES

- Failed-login validation
- Access-control enforcement
- Security testing

Authenticated Wireless Client Connected

VERIFIED



PROJECT EVIDENCE

Completed the lab with the RADIUS-authenticated laptop connected wirelessly to WAP1 while the server, switch, router, and administrator PC remained operational.

WHAT THIS DEMONSTRATES

- Secure wireless integration
- Centralized authentication outcome
- Topology validation

Technical Outcomes and Skills

CONFIGURATION OUTCOMES

- Built a centralized RADIUS identity service in Cisco Packet Tracer
- Configured WPA2-Enterprise wireless access using PEAP
- Registered network clients and created a centralized user identity
- Validated authorized access and rejected unauthorized authentication
- Confirmed connectivity to the access point, server, router, and administrator PC

SKILLS DEMONSTRATED

- Identity and Access Management
- RADIUS
- WPA2-Enterprise
- PEAP
- Cisco Packet Tracer
- Wireless security
- IPv4 addressing
- Network troubleshooting

PROJECT OUTCOME

This project demonstrates the ability to design and validate a centralized identity service, apply enterprise wireless authentication, protect credentials in portfolio evidence, test both successful and failed access attempts, and verify end-to-end network connectivity.

PROJECT STATUS

Complete - verified, documented, and prepared for GitHub portfolio publication.